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## Rock Mechanics (MIN E – 620)

Homework No. 2

### 1 Problem

A 6-ft coal seam at a depth of 750 ft is to be mined by a room-and-pillar system incorporating 16-ft entries and pillar and pillars that are 50 x 50 ft square. What must be the minimum laboratory compressive strength analysis for 4-in. cubes if stability is to be achieved? Use Holland-Gaddy formula.

### 2 Solution

The schematic sketch of the problem description is shown in the Figure 1.

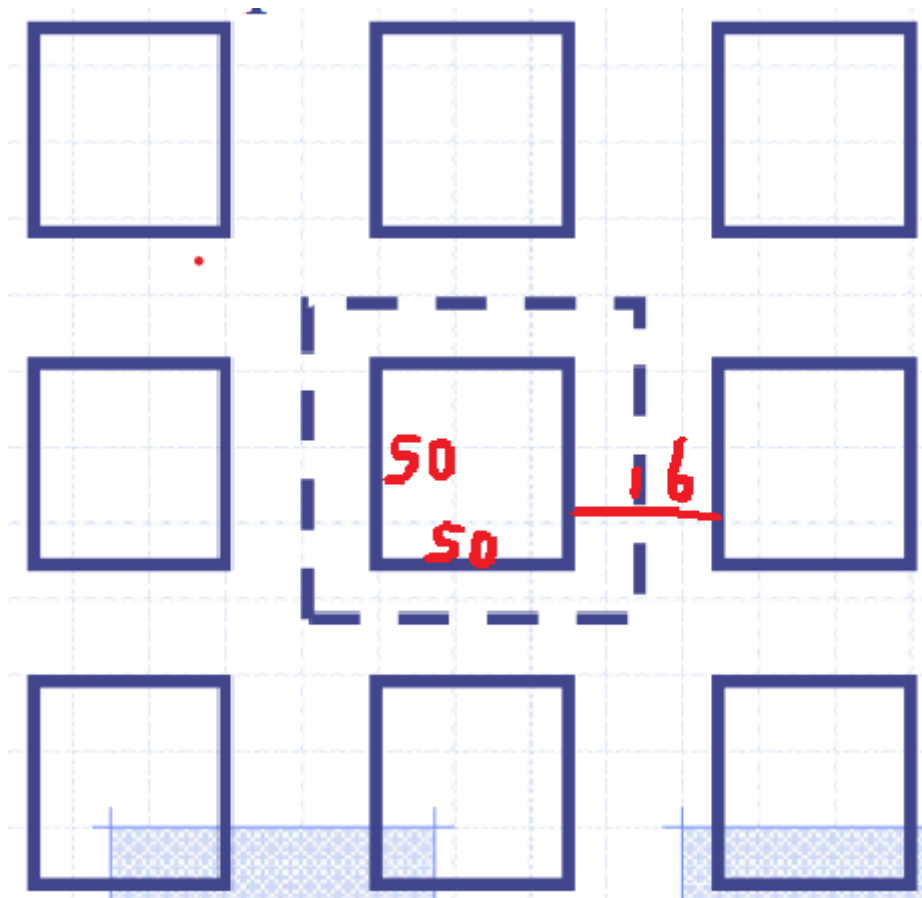


Figure 1: The schematic sketch of the problem description

Based on this, We have:

Area Total = Area of pillar + Area of entry=(50+16)ft x (50+16)ft=66ft x 66ft=4356 ft<sup>2</sup>

Pillar Area=50ft x 50ft=2500 ft<sup>2</sup>

Mined Out Area=Area Total - Pillar Area=4356 ft<sup>2</sup> - 2500 ft<sup>2</sup>=1856 ft<sup>2</sup>

Ratio of pillar area to total area (R)=1856 / 4356 =0.4264

The Holland-Gaddy formula is as shown in Equation 1:

$$\sigma_{pstrength} = \frac{KW^{0.5}}{H} \quad (1)$$

Where:

- $\sigma_{pstrength}$  ultimate compressive strength of pillar material, in psi
- $K$  coefficient depending on the coal comprising the pillar.
- $W$  least width of pillar, in inches
- $H$  thickness of pillar, in inches.

We use Equation 2 to estimate in-situ vertical stress:

$$\sigma_v \approx 1.1\text{psi/ft of depth} = 1.1 \times 750 = 825 \text{ psi} \quad (2)$$

Then, estimate the pillar stress using tributary area theory:

$$\sigma_p = \sigma_v \times \frac{1}{1-R} = 825 \times \frac{1}{1-0.4264} = 1438 \text{ psi}$$

Set  $\sigma_p = \sigma_{pstrength}$ , for make sure the minimum stability requirement.

$$1438 = \frac{K \times 600^{0.5}}{72} \implies K = 4226.8\text{psi}$$

The K is defined as the coefficient depending on the coal comprising the pillar, which can be estimated by the strength of the cube (lab specimen) using the Equation 3:

$$K = \sigma_c D^{0.5} \quad (3)$$

Where:

- $\sigma_c$  strength of the cube (lab specimen)
- $D$  edge dimension of the cube, in inches.

Substitute K=4226.8 and D=4 ft into Equation 3,

$$4226.8 = \sigma_c \times 4^{0.5} \implies \sigma_c = 2113.4\text{psi}$$

Thus, the minimum laboratory compressive strength for 4 inches must be greater than 2113.4 psi.